

PROFESSIONAL EXPERIENCE

Software Engineering Intern – AI Engineer Collège de Maisonneuve Research Lab

Montreal, Canada
August 2024 – November 2024

- Led a 4-person engineering team to design and ship an AI-driven educational platform for students with reading, writing, and comprehension difficulties.
- Engineered an accessibility-focused UI in React, optimizing component rendering for low-latency interactions.
- Integrated conversational LLMs to evaluate student essay plans, delivering structured, multi-turn feedback on writing and idea organization.
- Built CI/CD pipelines (GitHub Actions) automating testing, linting, and deployment across environments.

AI Developer & Project Coordinator Phon.ia – AI-Driven Phonetics & Literacy Platform

Montreal, Canada
November 2024 – May 2025

- Built a decoupled full-stack clinical speech platform (React + NestJS) serving both diagnostics and student tutoring.
- Designed dual-role JWT authentication with secure httpOnly refresh cookies, mitigating XSS/CSRF risk.
- Engineered an audio pipeline (Dockerized PostgreSQL + Ffmpeg) standardizing clips to 16kHz WAV for analysis.
- Developed a text-speech alignment algorithm detecting pronunciation errors (substitutions, omissions, inversions), visualized via custom Draft.js components.
- Integrated GPT-4o structured JSON schemas to automate clinical narrative assessment and session reports.

PROJECTS

Orbital – Autonomous AI Browser Agent

May 2026 – (present)

- Architected an autonomous browser agent converting natural language into real browser actions—navigation, form completion, and multi-step workflows—via **GPT-4o**.
- Built a perception-planning-execution loop that analyzes live page state and adapts to failures until task completion.
- Developed a semantic DOM engine compressing pages via Shadow DOM traversal and deterministic identifiers, cutting token usage.
- Shipped a **Manifest V3** Chrome Extension with a persistent Side Panel and **SSE** streaming for long-running execution; optimized inference cost via hierarchical model routing (GPT-4o-mini / GPT-4o).

Stack: React, TypeScript, Node.js, Express, Chrome APIs, OpenAI GPT-4o

PDFAnalyzerAI – Enterprise Document Intelligence Platform

Nov 2025 – Feb 2026

- Built a multi-tenant **RAG** platform transforming PDFs into searchable knowledge bases with semantic vector search.
- Engineered a document pipeline: parsing, chunking, OpenAI embeddings, **FAISS** indexing, GPT-4 contextual reasoning.
- Architected tenant isolation via per-user database separation and isolated vector indexes, preventing cross-tenant data leaks.
- Implemented deterministic citation retrieval, mapping AI responses directly to source pages in an integrated PDF viewer.

Stack: Python, Flask, PostgreSQL, FAISS, LangChain, GPT-4, React, Docker

SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL

Frontend / Backend: React, Next.js, Node.js, Express, NestJS, Flask

AI: OpenAI API, RAG, LangChain, Vector Search, Prompt Engineering

Infrastructure: Docker, PostgreSQL, FAISS, Git, GitHub Actions, CI/CD, JWT, REST APIs